

Ovidiu-Neculai Avadanei

Curriculum Vitae

California Institute of Technology
Division of Physics, Mathematics and
Astronomy
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Date of birth 01/19/1997

Appointments

September 2026-present: *Olga Taussky and John Todd Postdoctoral Scholar Teaching Fellow in Mathematics*, California Institute of Technology, United States of America

July 2025-June 2026: *Gibbs Assistant Professor*, Yale University, United States of America

Education

August 2019-May 2025: PhD in Mathematics, University of California at Berkeley, United States of America, GPA 4.0

Doctoral Advisor: Professor Daniel Tataru

October 2016-July 2019: Bachelor's Degree in Mathematics, "Alexandru Ioan Cuza" University of Iasi, Romania (I graduated with a perfect GPA in a 10 point system, as valedictorian)

Research Interests

Nonlinear dispersive partial differential equations, fluid dynamics, mathematical physics

Publications and Preprints

1. O. Avadanei, W. Schlag, *Rigorous analysis of giant magnetic vortex strings*, arXiv preprint, arXiv:2609.194
2. O. Avadanei, *A conditional Lagrangian clock barrier at the $C^{1,\frac{1}{3}}$ threshold for axisymmetric Euler without swirl*, arXiv preprint, arXiv:2605.31587
3. O. Avadanei, *Counterexamples to Strichartz estimates and gallery waves for the irrotational compressible Euler equation in a vacuum setting*, arXiv preprint, arXiv:2504.17932
4. A. Ai, O. Avadanei, *Low regularity well-posedness for the generalized surface quasi-geostrophic front equation*, Pure and Applied Analysis, Volume 8, Number 2, 2026, Pages 271-329. The article can be found [here](#).
5. A. Ai, O. Avadanei, *Low regularity solutions for the surface quasi-geostrophic front equation*, arXiv preprint, arXiv:2310.20143
6. A. Ai, O. Avadanei, *Well-posedness for the surface quasi-geostrophic front equation*, **Nonlinearity**, Volume 37, Number 5, May 2024. The article can be found [here](#).

7. A. Ai, O. Avadanei, *Well-posedness for the dispersive Hunter-Saxton equation*, ***International Mathematics Research Notices***, Volume 2023, Issue 9, May 2023, Pages 7845-7882. The article can be found [here](#).

Conferences, Workshops, Summer Schools, and Presentations

Invited Talks

- ***Recent Trends in Fluid Dynamics***, Brin Mathematics Research Center, University of Maryland, College Park, Maryland, USA, April 9, 2026

Contributed Talks

1. ***The 21st Prairie Analysis Seminar***, Kansas State University, Manhattan, Kansas, USA, November 7-8, 2025
2. ***Brown-Yale Analysis/PDE Seminar***, Brown University, Providence, Rhode Island, USA, October 25, 2025
3. ***Conference on Mathematics of Wave Phenomena***, Karlsruhe Institute of Technology, Karlsruhe, Germany, February 24, 2025 (virtual talk)
4. ***Workshop for Young Researchers in Mathematics***, "Alexandru Ioan Cuza" University of Iasi, Iasi, Romania, May 23, 2024
5. ***Nonlinear Waves and Relativity Thematic Programme***, Erwin Schrödinger Institute, Vienna, Austria, May 14, 2024
6. ***AMS Spring Central Sectional Meeting***, University of Wisconsin-Milwaukee, Milwaukee, Wisconsin, USA, April 20-21, 2024
7. ***Spring School on Soliton Dynamics***, Texas A&M University, College Station, Texas, USA, March 14-16, 2024
8. ***The 19th Prairie Analysis Seminar***, Kansas State University, Manhattan, Kansas, USA, November 3-4, 2023
9. ***Mathematical Problems in Fluid Dynamics, part 2***, Simons Laufer Mathematical Sciences Institute, Berkeley, California, USA, July 17-August 11, 2023
10. ***The Tenth Congress of Romanian Mathematicians***, University of Pitești, Pitești, Romania, June 30-July 5, 2023
11. ***Shanks Workshop on Mathematical Aspects of Fluid Dynamics***, Vanderbilt University, Nashville, Tennessee, USA, February 19-20, 2022

Seminar Talks

1. *Harmonic Analysis and PDE Student Seminar*, University of California at Berkeley, Berkeley, California, USA, September 15, 2026
2. *Analysis and PDE Seminar*, University of California at Berkeley, Berkeley, California, USA, September 14, 2026
3. *PDE Seminar*, Brown University, Providence, Rhode Island, USA, March 20, 2026
4. *PDE Seminar*, Pennsylvania State University, State College, Pennsylvania, USA, February 18, 2026
5. *Analysis Seminar*, Yale University, New Haven, Connecticut, USA, April 28, 2025
6. *Analysis and PDE Seminar*, University of California at Berkeley, Berkeley, California, USA, April 21, 2025
7. *Analysis Seminar*, California Institute of Technology, Pasadena, California, USA, November 27, 2024
8. *USC Analysis and PDE Seminar*, University of Southern California, Los Angeles, California, USA, November 25, 2024
9. *Applied and Computational Mathematics Student Seminar*, University of South Carolina, Columbia, South Carolina, USA, November 11, 2024 (virtual talk)
10. *Analysis Seminar*, Columbia University, New York, New York, USA, September 20, 2024
11. *Analysis and PDE Seminar*, University of California at Los Angeles, Los Angeles, California, USA, April 30, 2024
12. *Harmonic Analysis and PDE Student Seminar*, University of California at Berkeley, Berkeley, California, USA, September 19, 2023
13. *Harmonic Analysis and PDE Student Seminar*, University of California at Berkeley, Berkeley, California, USA, March 7, 2023
14. *PDE and Geometric Analysis Seminar*, University of Wisconsin-Madison, Madison, Wisconsin, USA, February 20, 2023
15. *Harmonic Analysis and PDE Student Seminar*, UC Berkeley, Berkeley, California, USA, October 12, 2021
16. *Graduate Student Working Group*, as part of the *Mathematical problems in fluid dynamics(Virtual)* program, Mathematical Sciences Research Institute, Berkeley, California, May 5, 2021
17. *Sesiunea Nationala de Comunicari Stiintifice Studentesti de Matematica*, "Alexandru Ioan Cuza" University of Iasi, Romania, July 4-7, 2019

Poster presentations

1. *Rivière-Fabes Symposium on Analysis and PDE*, University of Minnesota-Twin Cities, Minneapolis, Minnesota, USA, April 25-27, 2025

Conferences attended as participant

1. *International Congress of Mathematicians*, Philadelphia, Pennsylvania, USA, July 23-30, 2026
2. *Nonlinear Waves and Dispersive Equations Workshop*, Mathematisches Forschungsinstitut Oberwolfach, Oberwolfach-Walke, Germany, March 1-6, 2026
3. *3rd Simons Math Summer Workshop: Partial Differential Equations of Classical Physics*, Simons Center for Geometry and Physics, Stony Brook, New York, USA, July 10-18, 2025
4. *Paris-Saclay conference in Analysis and PDE*, Laboratoire de Mathématiques d'Orsay, Orsay, France, May 27-31, 2024
5. *Recent developments in wave turbulence theory*, University of North Carolina at Chapel Hill, Chapel Hill, North Carolina, USA, April 5-6, 2024
6. *Advances in Nonlinear Analysis and Nonlinear Waves*, IHES, Le Bois-Marie, France, May 22-26, 2023
7. *Southern California Analysis and Partial Differential Equations*, University of California at San Diego, San Diego, California, USA, November 2-3, 2019
8. *Instruments of Algebraic Geometry*, University of Bucharest, Faculty of Mathematics and Computer Science, September 17-22, 2017, Bucharest, Romania
9. *XIIIème Colloque Franco-Roumain de Mathématiques Appliquées*, "Alexandru Ioan Cuza" University of Iasi, Faculty of Mathematics, August 25-29, 2016

Summer Schools and Workshops

1. *Oberwolfach Seminar: Free Boundary Problems in Fluid Dynamics*, Mathematisches Forschungsinstitut Oberwolfach, Oberwolfach-Walke, Germany, October 23-29, 2022
2. *Shanks Workshop on Mathematical Aspects of Fluid Dynamics*, Vanderbilt University, Nashville, Tennessee, February 19-20, 2022
3. *Summer Program in Partial Differential Equations (Virtual Summer Graduate School)*, University of Texas at Austin, Austin, May 17-21, 2021 and May 24-28, 2021
4. *Introductory Workshop: Mathematical problems in fluid dynamics (Virtual Workshop)*, Mathematical Sciences Research Institute, Berkeley, January 25-February 05, 2021

5. *Connections Workshop: Mathematical problems in fluid dynamics (Virtual Workshop)*, Mathematical Sciences Research Institute, Berkeley, January 20-22, 2021
6. *Mathematical problems in fluid dynamics (Virtual Program)*, Mathematical Sciences Research Institute, Berkeley, January 19-May 28, 2021
7. *Introduction to water waves (Virtual Summer Graduate School)*, Mathematical Sciences Research Institute, Berkeley, July 27-August 07, 2020
8. *Sesiunea Nationala de Comunicari Stiintifice Studentesti de Matematica*, "Alexandru Ioan Cuza" University of Iasi, July 4-7, 2019
9. *Scuola Matematica Interuniversitaria*, Universita degli Studi di Perugia, Department of Mathematics and Computer Science, July 23-August 17, 2018, where I attended the "*Commutative Algebra and Geometry*" and "*Introduction to Mathematical Statistics*" courses
10. *Workshop for Young Researchers in Mathematics*, "Simion Stoilow" Institute of Mathematics of the Romanian Academy (IMAR), May 17-18, 2018
11. *Instruments of Algebraic Geometry*, University of Bucharest, Faculty of Mathematics and Computer Science, September 11-15, 2017, Bucharest, Romania
12. *Modern Mathematics Summer School*, Jacobs University, Bremen, Germany, July 1-11, 2017, where we were given introductory lectures in many fields of Mathematics, such as Representation Theory, Modular Forms, and Probability Theory

Academic Services

- Referee for *Annals of PDE*, *Inventiones Mathematicae*, and *Communications in Analysis and Geometry*
- Co-organizer of the "Recent developments in the dynamics of nonlinear dispersive equations I" Special Session as part of the AMS Spring Central Sectional Meeting, April 18-19, 2026
- Co-organizer of the UC Berkeley *Harmonic Analysis and Differential Equations Student Seminar (HADES)* (Fall 2023-Spring 2024)

Selected Awards and Grants

Awards

1. Herb Alexander Prize for outstanding dissertation in Pure Mathematics, 2024-2025
2. Silver Medal at the 57th *International Mathematical Olympiad*, July 6-16, 2016, Hong Kong
3. First Prize at the 24th *International Mathematics Competition for University Students*, July 31-August 7, 2017, Blagoevgrad, Bulgaria
4. Silver Medal at the 33rd *Balkan Mathematical Olympiad*, May 5-10, 2016, Tirana, Albania

5. **Gold Medal and Absolute Champion** at the *South Eastern European Mathematical Olympiad for University Students 2018*, February 27-March 4, 2018, Iasi, Romania
6. **Gold Medal** at *South Eastern European Mathematical Olympiad for University Students 2017*, February 28-March 5, 2017, Ohrid, Macedonia
7. **Bronze Medal** at the 8th *"Romanian Masters of Mathematics" International Competition*, February 24-29, 2016
8. **Silver Medal** at the 16th *Junior Balkan Mathematical Olympiad, June 25-29, 2012, Veria, Greece*
9. **Gold Medals** at the *Romanian National Mathematical Olympiad* in 2015 and 2016
10. **First Prize** at the *"Traian Lalescu" National Mathematical Competition for University Students*, May 5-7, 2017, Constanta, Romania

Grants

Fellowship for Young Researchers, awarded by the Romanian Ministry of National Education, 2016-2018

Teaching Experience

1. **Gibbs Assistant Professor.** Department of Mathematics, Yale University, New Haven, Connecticut, United States of America. Fall 2025.
2. **Graduate Student Instructor.** Department of Mathematics, University of California at Berkeley, Berkeley, California, United States of America. Fall 2019-Fall 2021, Fall 2022, Spring 2024-Spring 2025.

Internships

February 4, 2017- September 4, 2017: S.C. Gemini CAD Systems, Iasi

Project: CUDA Nesting (a topic in Computational Geometry regarding the irregular stock cutting problem), under the supervision of Mr. Radu Danciu (Software Development Manager)

Miscellaneous

1. **Coordinator (grader)** at the *59th International Mathematical Olympiad*, Cluj-Napoca, Romania, July 9-14, 2018
2. **Member of the Central Committee (grader)** at the *National Mathematical Olympiad*, Timisoara, Romania, April 19-23, 2017

Other skills

Languages

Romanian (native), English (fluent), German (intermediate)

Computer skills

Working knowledge of C++, MATLAB, Python, L^AT_EX, as well as familiarity with the Microsoft Office Suite